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Assignment 0: Study of Basic Linux Commands

Aim:-

The aim of this assignment is to study and understand basic Linux commands used for file handling, general system operations, shell usage, networking, and process management. This assignment also aims to practice these commands along with their various options by referring to the man command, in order to gain hands-on experience and improve command-line efficiency in the Linux operating system.

Objective:

1. To know the basic Linux commands related to:
    a. File handling
    b. General
    c. Shell
    d. Networking
    e. Process management etc....
2. To practice this commands with its various options by through “man” command

Theory:-

Linux Kernel–Based Operating Systems

Linux kernel–based operating systems are OSs that use the Linux kernel as their core. The kernel is the heart of the system—it manages hardware, memory, processes, files, and device drivers. Linux itself is not a complete operating system; it becomes one only when combined with system libraries, utilities, shells, and user interfaces. These OSs are open-source, secure, stable, and widely used in servers, desktops, mobile devices, embedded systems, and cloud platforms.

Linux Distributions

A Linux distribution (distro) is a complete operating system built around the Linux kernel, along with system tools, package managers, desktop environments, and applications. Different distributions exist to serve different needs such as beginners, developers, servers, security professionals, or enterprise use. Popular Linux distributions include Ubuntu, Debian, Fedora, Red Hat Enterprise Linux (RHEL), Arch Linux, Kali Linux, and Linux Mint.

Types of Linux Distributions

Linux distributions can be broadly categorized based on their target users and purpose. Desktop distributions like Ubuntu and Linux Mint focus on ease of use. Server distributions like RHEL, CentOS Stream, and Ubuntu Server focus on stability and long-term support. Rolling-release distributions like Arch Linux continuously update software instead of fixed releases. Security-focused distributions like Kali Linux are designed for penetration testing and cybersecurity tasks.

Linux Kernel Versions

Linux kernel versions follow a numeric format such as 6.1, 6.6, etc. Earlier versions used a three-number scheme (for example, 3.10.x), but modern kernels follow a simpler versioning model. New kernel versions introduce performance improvements, hardware support, security patches, and new features. Some kernel versions are marked as LTS (Long-Term Support), meaning they receive updates and security fixes for many years.

Kernel Version vs Distribution

The kernel version is independent of the Linux distribution. Multiple distributions can use the same kernel version, and the same distribution can ship with different kernel versions over time. For example, Ubuntu 22.04 may ship with kernel 5.15 by default, while allowing users to upgrade to newer kernels. This separation allows flexibility and customization based on system requirements.

Distribution Versions

Distribution versions represent releases of a Linux distribution and usually include a specific kernel version, system libraries, and software packages. These versions are identified using numbers or code names, such as Ubuntu 20.04 (Focal Fossa) or Fedora 39. Distribution versions determine software compatibility, support duration, and update cycles. LTS distribution versions are preferred in production environments due to long-term stability.

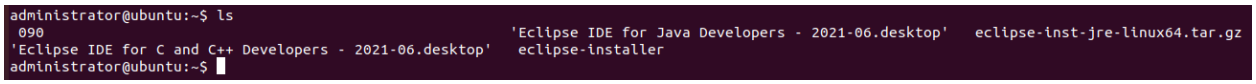
Relationship Between Kernel and Distribution Versions

A distribution version bundles a tested kernel version along with compatible software. While the kernel evolves rapidly, distributions choose stable kernel versions to ensure reliability. Users can update kernels without changing the distribution version, but upgrading the distribution version typically includes a newer kernel and updated system components.

Commands:-

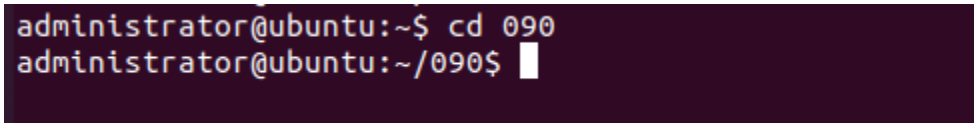
1.Ls

The `ls` command is one of the most commonly used Linux commands and is used to list the contents of a directory. By default, it displays the names of files and subdirectories present in the current working directory.



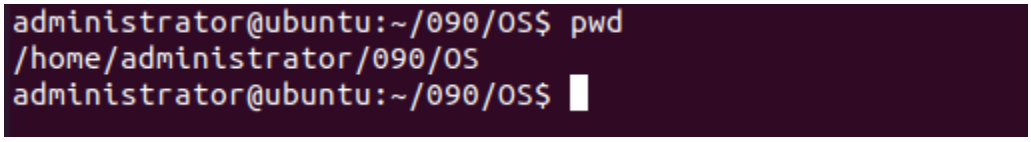
2. Cd

The `cd` (change directory) command is used to move from one directory to another in the Linux file system. It helps users navigate folders easily by specifying a directory path or using shortcuts like `..` (parent directory) and ~ (home directory).



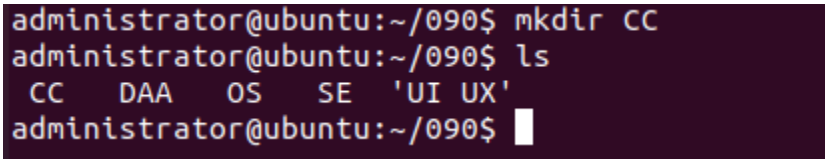
3.pwd

The `pwd` (print working directory) command is used to display the full path of the current directory in which the user is working. It helps users know their exact location in the Linux file system.



4.mkdir

The `mkdir` (make directory) command is used to create new directories in the Linux file system. It allows users to organize files by creating folders, and with options it can also create multiple directories or nested directories at once.



5.rmdir

The `rmdir` (remove directory) command is used to delete empty directories from the file system. It only works if the directory contains no files or subdirectories, helping prevent accidental data loss.

```
administrator@ubuntu:~/090$ rmdir CC
administrator@ubuntu:~/090$ ls
 DAA   OS   SE  'UI UX'
administrator@ubuntu:~/090$
```

6.rm

The `rm` (remove) command is used to delete files and directories from the Linux file system. It permanently removes data, so it should be used carefully. With options, it can remove multiple files or entire directories.

```
administrator@ubuntu:~/090/SE$ rm file.txt
administrator@ubuntu:~/090/SE$ ls
administrator@ubuntu:~/090/SE$
```

7.touch

The `touch` command is used to create an empty file in Linux. It can also be used to update the access and modification time of an existing file without changing its content.

```
administrator@ubuntu:~/090$ cd OS
administrator@ubuntu:~/090/OS$ touch file.txt
administrator@ubuntu:~/090/OS$ cat file.txt
Hello World!

administrator@ubuntu:~/090/OS$
```

8.cat

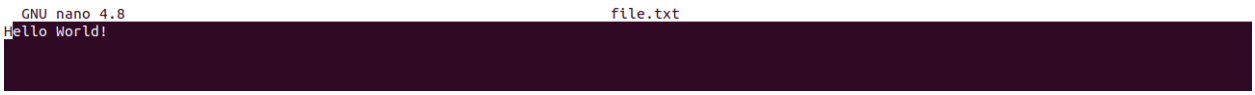
The `cat` (concatenate) command is used to display the contents of a file on the terminal. It can also be used to combine multiple files or create a file by redirecting output.

```
administrator@ubuntu:~/090$ cd OS
administrator@ubuntu:~/090/OS$ touch file.txt
administrator@ubuntu:~/090/OS$ cat file.txt
Hello World!

administrator@ubuntu:~/090/OS$
```

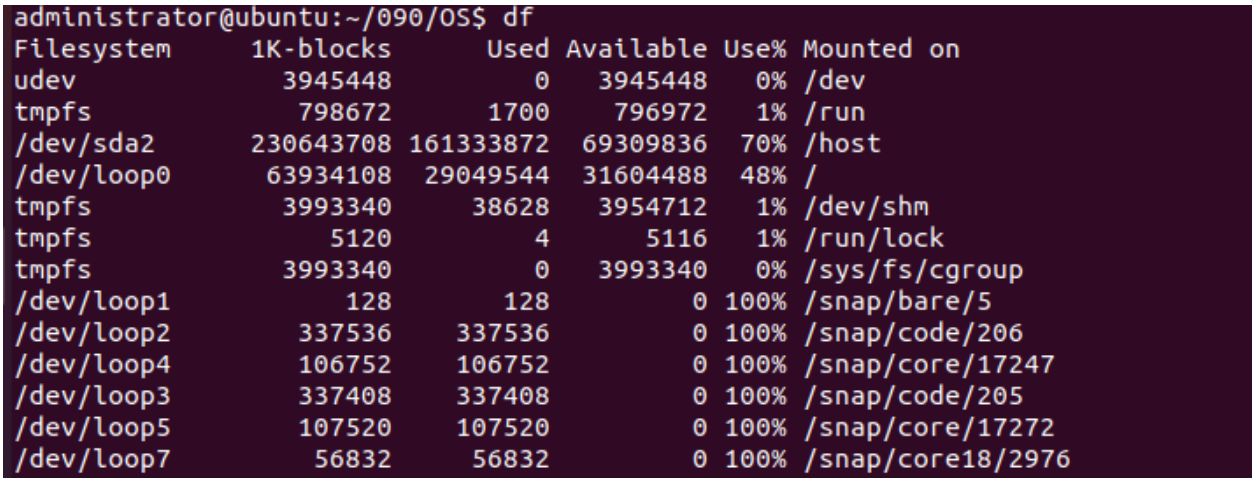
9.nano:-

The nano command is a simple, user-friendly text editor used in the Linux terminal. It allows users to create and edit text files directly from the command line, with basic shortcuts displayed at the bottom of the editor.



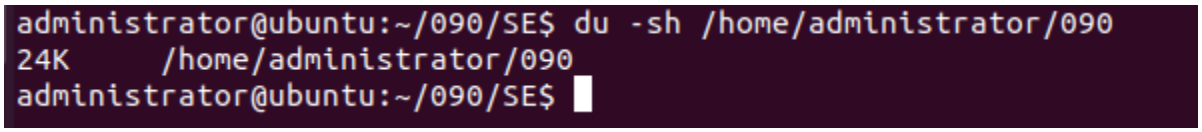
10.df:-

The df (disk free) command is used to display information about disk space usage of file systems. It shows total space, used space, available space, and mount points, helping users monitor storage availability.



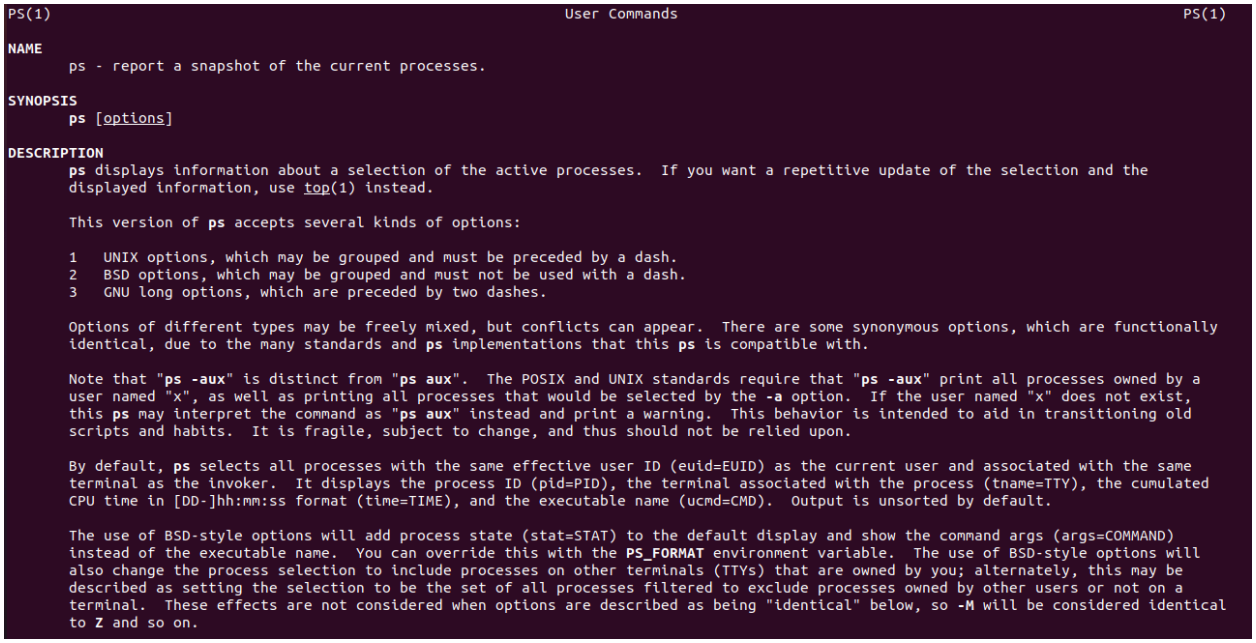
11.du:-

The du (disk usage) command is used to display the disk space used by files and directories. It helps users identify how much space individual directories or files are consuming.



12.ps

The ps (process status) command is used to display information about currently running processes in the system. It shows details such as process ID (PID), terminal, CPU usage, and command name.



PS(1) User Commands PS(1)

NAME
ps - report a snapshot of the current processes.

SYNOPSIS
ps [options]

DESCRIPTION
ps displays information about a selection of the active processes. If you want a repetitive update of the selection and the displayed information, use top(1) instead.

This version of **ps** accepts several kinds of options:

1 UNIX options, which may be grouped and must be preceded by a dash.
2 BSD options, which may be grouped and must not be used with a dash.
3 GNU long options, which are preceded by two dashes.

Options of different types may be freely mixed, but conflicts can appear. There are some synonymous options, which are functionally identical, due to the many standards and **ps** implementations that this **ps** is compatible with.

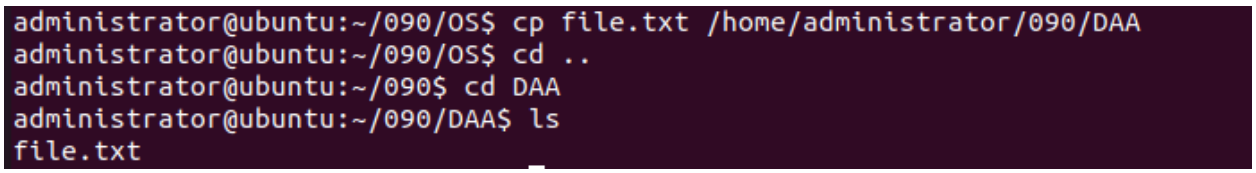
Note that "**ps -aux**" is distinct from "**ps aux**". The POSIX and UNIX standards require that "**ps -aux**" print all processes owned by a user named "x", as well as printing all processes that would be selected by the **-a** option. If the user named "x" does not exist, this **ps** may interpret the command as "**ps aux**" instead and print a warning. This behavior is intended to aid in transitioning old scripts and habits. It is fragile, subject to change, and thus should not be relied upon.

By default, **ps** selects all processes with the same effective user ID (euid=EUID) as the current user and associated with the same terminal as the invoker. It displays the process ID (pid=PID), the terminal associated with the process (tname=TTY), the cumulated CPU time in [DD-]hh:mm:ss format (time=TIME), and the executable name (ucmd=CMD). Output is unsorted by default.

The use of BSD-style options will add process state (stat=STAT) to the default display and show the command args (args=COMMAND) instead of the executable name. You can override this with the **PS_FORMAT** environment variable. The use of BSD-style options will also change the process selection to include processes on other terminals (TTYs) that are owned by you; alternately, this may be described as setting the selection to be the set of all processes filtered to exclude processes owned by other users or not on a terminal. These effects are not considered when options are described as being "identical" below, so **-M** will be considered identical to **Z** and so on.

13. cp:-

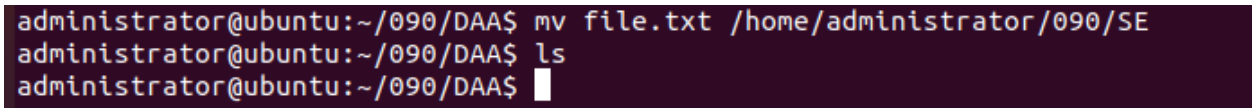
The cp (copy) command is used to copy files or directories from one location to another. It allows users to duplicate data while keeping the original file unchanged.



```
administrator@ubuntu:~/090/OS$ cp file.txt /home/administrator/090/DAA
administrator@ubuntu:~/090/OS$ cd ..
administrator@ubuntu:~/090$ cd DAA
administrator@ubuntu:~/090/DAA$ ls
file.txt
```

14.mv:-

The mv (move) command is used to move files or directories from one location to another. It is also commonly used to rename files and directories in Linux.



```
administrator@ubuntu:~/090/DAA$ mv file.txt /home/administrator/090/SE
administrator@ubuntu:~/090/DAA$ ls
administrator@ubuntu:~/090/DAA$
```

15.find:-

The `find` command is used to search for files and directories in the Linux file system based on different criteria such as name, type, size, or modification time. It is very useful for locating files in large directory structures.

```
administrator@ubuntu:~/090$ cd OS
administrator@ubuntu:~/090/OS$ find . -name "file.txt"
./file.txt
administrator@ubuntu:~/090/OS$
```

16.grep:-

The `grep` command is used to search for specific text or patterns within files. It scans file contents line by line and displays the lines that match the given pattern, making it helpful for log analysis and text searching.

```
administrator@ubuntu:~/090/OS$ grep "pattern" file.txt
administrator@ubuntu:~/090/OS$
```

17.chmod:-

Working:-Change file permissions.

```
administrator@ubuntu:~/090/OS$ chmod 755 file.txt
administrator@ubuntu:~/090/OS$
```

18.top:-

Working:- Dynamic process viewer.

```
top - 10:23:05 up  1:17,  1 user,  load average: 0.38, 0.62, 0.69
Tasks: 269 total,   1 running, 268 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.5 us,  0.5 sy,  0.0 ni, 98.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7799.5 total,    266.8 free,   2134.1 used,   5398.6 buff/cache
MiB Swap:    256.0 total,    245.9 free,     10.1 used.   4993.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1909 adminis+  20   0 4485884 274380 101788 S  12.0   3.4   4:44.30 gnome-shell
   1782 adminis+  20   0  585136  92564  56872 S   3.3   1.2   2:04.61 Xorg
   1152 mysql     20   0 2381224 396712  36576 S   1.0   5.0   0:44.22 mysqld
     67 root      20   0       0      0      0 S   0.3   0.0   0:04.09 kcompactd0
    158 root       0 -20       0      0      0 I   0.3   0.0   0:04.28 kworker/u17:0-i915_flip
    264 root      20   0    3748   2080   1792 S   0.3   0.0   0:31.64 mount.ntfs
   1016 mongodb   20   0  979992  73580  32028 S   0.3   0.9   0:22.47 mongod
```

19.htop:-

Working:- Enhanced version of top.

```
administrator@ubuntu:~/090$ htop

Command 'htop' not found, but can be installed with:

sudo snap install htop  # version 3.4.1, or
sudo apt  install htop  # version 2.2.0-2build1

See 'snap info htop' for additional versions.
```

20.free:-

The `free` command is used to display information about system memory usage. It shows total, used, and available RAM and swap memory, helping users monitor memory consumption in the Linux system.

```
administrator@ubuntu:~/090/OS$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       2.1Gi       194Mi       404Mi       5.3Gi       4.8Gi
Swap:          255Mi        10Mi       245Mi
administrator@ubuntu:~/090/OS$
```